Date: 15/10

Class: XII Boards

500

72%

72% of 500

$\frac{72}{100} \times 500$

$\frac{100}{600} \times 100$

96%

Fail = 4%

minus = 2272

Let total candidates = x

4% of x = 2272

$\frac{4}{100} \times x = 2272$

$x = \frac{2272 \times 100}{4}$

Date: 15/10

Class: X

5. Given : In trapezium PQRS, side PQ || side SR, AR = 5AP, AS = 5AQ then prove that, SR = 5PQ

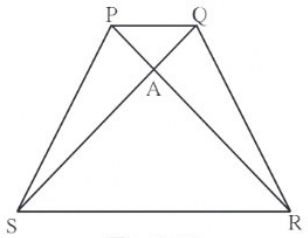


Fig. 1.59

$AR = 5AP$

$\frac{AR}{AP} = 5$

$\frac{AS}{AQ} = 5$

$\triangle PSR, \triangle QSR$

$SR = SR$

$\frac{AR}{AP} = \frac{AS}{AQ} = 5$

$\triangle PSR \sim \triangle QSR$ {SSS test}

$\frac{SR}{PQ} = 5 \quad SR = 5PQ$

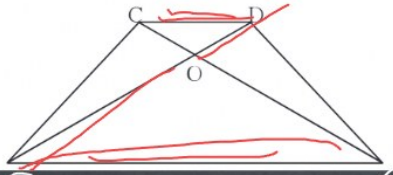


Fig. 1.59

6. In trapezium ABCD, (Figure 1.60) side AB || side DC, diagonals AC and BD intersect in point O. If AB = 20, DC = 6, OB = 15 then find OD.

$\angle COD = \angle BOA$ { Opp angles }

$\angle DCO = \angle OAB$

$\angle CDO = \angle OBA$ { Alternate angle }

AAA

$\triangle COD \sim \triangle AOB$

$\frac{AB}{CD} = \frac{OB}{OD}$

$\frac{20}{6} = \frac{15}{OD}$

$20 \times OD = 15 \times 6$

$OD = \frac{15 \times 6}{20}$

Date: 15/10

Class: X

7. ABCD is a parallelogram point E is on side BC. Line DE intersects ray AB in point T. Prove that $DE \times BE = CE \times TE$.

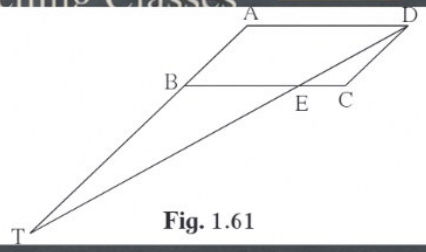


Fig. 1.61

$\angle BET = \angle CED$ {Opp $\angle$}

$\angle TBE = \angle ECD$ {Alternate $\angle$}

$\triangle CED \sim \triangle BET$ {AA test}

$$\frac{BE}{CE} = \frac{TE}{DE}$$

Date: 15/10

Class: X

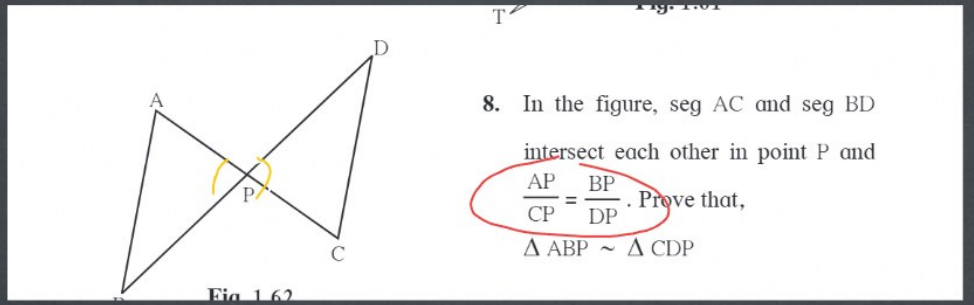


Fig. 1.62

8. In the figure, seg AC and seg BD intersect each other in point P and $\frac{AP}{CP} = \frac{BP}{DP}$. Prove that,

$\Delta ABP \sim \Delta CDP$

$\frac{AP}{CP} = \frac{BP}{DP}$

$\angle APB = \angle CPD$ {opp}

$\triangle ABP \sim \triangle CDP$ {SAS test}

Date: 15/10

Class: X

9. In the figure, in Δ ABC, point D on side BC is such that,

$\angle$ BAC = $\angle$ ADC.

Prove that, $CA^2 = CB \times CD$

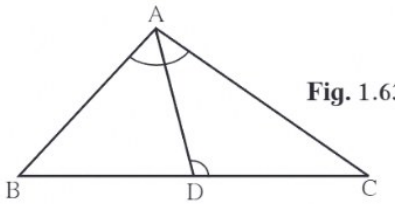


Fig. 1.63

In $\triangle BAC$ & $\triangle ADC$

$\angle BAC = \angle ADC$

$\angle ACB = \angle ACD$

$\triangle BAC \sim \triangle ADC$ {AA test}

$$\frac{AC}{CD} = \frac{CB}{AC}$$